

# CELIA RUBIO-MADRIGAL

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I work on graph neural network generalization and efficiency, with an emphasis on understanding and evaluating model limitations. Background in mathematics and computer science; “la Caixa” Fellow and HLF attendee.

## EXPERIENCE



- **Doctoral researcher** – *CISPA Helmholtz Center for Information Security* **2023-now**  
– Graph model generalization, efficiency, benchmarking. – Advisor: Dr. Rebekka Burkholz  
- **RA: Departmental collaboration** – *Ministerio de Educación & Universidad Complutense de Madrid* **2021-22**  
– “Identifying Boolean functions that need big circuits with ML”  – Advisor: Dr. Ismael Rodríguez Laguna 
- **REU: Introduction to research ICMAT-Severo Ochoa** – *Instituto de Ciencias Matemáticas (ICMAT)* **2021**  
– “Differential geometry methods in ML applications”  – Advisor: Dr. David Martín de Diego 
- **Intern: Consultant Developer** – *Guidewire Software* – NLP search engine to find internal raised issues. **2021**

## EDUCATION



- **PhD in Computer Science** – *CISPA & Universität des Saarlandes (Saarbrücken, Germany)* **2023-now**
- **MSc in Advanced CS with AI** – *University of Strathclyde (Glasgow, UK)* – 8.57/10 (Distinction) **2022-23**  
– Recipient of the “la Caixa” Fellowship  – Department’s Best Overall Performance Prize (Top 1)
- **BSc in Mathematics** – *Universidad Complutense de Madrid (Spain)* – 9.08/10 **2017-22**
- **BSc in Computer Science** – *Universidad Complutense de Madrid (Spain)* – 9.16/10 **2017-22**

## PUBLICATIONS



Legend: **bold**: First author in A\* conference. \* Equal contribution. † Co-supervision.

- R-M., Burkholz. *Linear GCNs Need Better Bias, Not More Expressive Power*. GFM Workshop @ ICML’26. 
- Joshi, Gadhikar<sup>†</sup>, R-M.<sup>†</sup>, Dutta, Singh, Burkholz. *Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning*. MemFM Workshop @ ICML’26. 
- R-M., Burkholz. *Fixed Aggregation Features Can Rival GNNs*. **ICML’26**. 
- Reddy, Verma, R-M., Muandet, Burkholz. *Boosting for Predictive Sufficiency*. ICLR’26. 
- Reddy, R-M., Burkholz, Muandet. *When Shift Happens - Confounding Is to Blame*. ICLR’26. 
- Jacobs, Gadhikar, R-M., Burkholz. *Hyperbolic Aware Minimization: Implicit Bias for Sparsity*. ICLR’26. 
- R-M.\*, Jamadandi\*, Burkholz. *GNNs Getting Comfy: Community and Feature Similarity Guided Rewiring*. **ICLR’25**. 
- Jamadandi\*, R-M.\*, Burkholz. *Spectral Graph Pruning against Over-squashing and Over-smoothing*. **NeurIPS’24**. 

## AWARDS



- 13th Heidelberg Laureate Forum: 200 international young researchers network with math and CS laureates **2026**
- Dept. Best Overall Performance Prize: *University of Strathclyde's top student in MSc Advanced CS* **2023**
-  Nova III Student List: *Spain's top 10 talents under 25 in Computer Science* **2023**

- 9th Heidelberg Laureate Forum. Featured in: 📺 *HLFF Spotlight*, 📺 *Tribuna Complutense* 2022
- 🏆 "la Caixa" Fellowship: *Full funding for MSc studies*. Featured in: 📺 *El País*, 📺 *El Plural*, 📺 *TeleMadrid* 2022
- **Undergraduate**
  - 🏆 Aula de Verano "Blas Cabrera" for Spain's top undergrad records: *Intl. Univ. Menéndez Pelayo (UIMP)* 2022
  - 🏆 SWERC contestant (competitive programming): *ICPC Southwestern Europe Regional Contest in Italy* 2022
  - Becas de Excelencia: *Excellence Grants for undergrads with >9/10 in Madrid (2100€)* 2018-22
- **Pre-undergraduate**
  - 🏆 Premio Extraordinario de Bachillerato: *Madrid's top 25 in general-content exam (1000€)* 2017
  - 🏆 EGMO contestant representing Spain: *European Girls' Mathematical Olympiad in Switzerland* 2017

## TALKS



- **Invited talks** – *Seminars and reading groups* 2025-26
  - 🏆 GLOW, 🏆 TUWien ML RU seminar, 📺 Cohere For AI, CISPA ML seminar, 🏆 AI Security Afternoon.
- **Contributed talks** – *Conferences and workshops* 2024-26
  - GSP 🏆 2026, HAICON 🏆 2024. Lightning talks: GTML 🏆 2025, MLG 🏆 2025 (2), NetSci 🏆 2025.
- **Non-archival posters** – *Non-archival conferences, workshops, and sessions* 2024-26
  - GSP 🏆 2026, CPAL 🏆 2026 (3), WiML 🏆 2024 🏆 2025, ELLIS Pre-NeurIPS/ICML 🏆 2024 🏆 2025 🏆 2026.

## TEACHING



- **Math Olympiad Trainer** – 🏆 *International EGMO Preparatory Seminar in Bonn, Germany* 2026
- **Master's and Bachelor's thesis advising** – *CISPA & Universität des Saarlandes (3)* 2024-now
- **Teaching Assistant: Laboratory demonstrator (7h/week)** – *University of Strathclyde* 2023
  - "CS310 Foundations of AI" for Year 3 BSc Computer Science – Lecturer: Dr. Andrew Abel 🏆

## SERVICE



- **Academic service** 2024-now
  - Reviewer: MLG 2025-26, ICLR 2026 – Top 200 (1%), CPAL 2026, ICML 2026 – Gold (25%), NeurIPS 2026.
  - Volunteer at 19th and 20th Women in Machine Learning Workshop (WiML) at NeurIPS 2024 and 2025.
- **Math Olympiad service**
  - 🏆 IT at European Girls' Mathematical Olympiad (EGMO) with Joseph Samuel Myers 🏆 2023-now
  - Deputy Leader of Spain at EGMO 2022 and PAGMO 2021. 2021-22
  - Jury at the local phases of 57th to 61st Olimpiada Matemática Española (OME) Madrid. 2020-24
  - Guide at the Iberoamerican Mathematical Olympiad 2018.
- **Student representation** – *Facultad de Informática (Fdi), Universidad Complutense de Madrid* 2020-22
  - Secretary of the Fdi Student Council 2021-22. Member of the Fdi Faculty Board 2020-22.
- **L<sup>A</sup>T<sub>E</sub>X package development: ddphonism** 2019
  - 🏆 Twelve-tone music notation diagrams with TikZ. Included in MiKTeX and TeXLive (try it in Overleaf).
- **Mentoring and outreach**
  - Mentor for new "la Caixa" Fellows 2023-25
  - Outreach articles for DivulgaMAT (RSME journal): *Serialismo y matemáticas* 📄 📄 📄 📄 2019
  - Outreach videos for the Spanish math education community: 📺 📺 📺 2018-21